Status	Finished
Started	Saturday, 17 October 2026, 6:18 PM
Completed	Saturday, 17 October 2026, 6:25 PM
Duration	7 mins 21 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Comparing each element with the others
c. Dividing the input into smaller subproblems
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By sorting data in a single pass
b. By using the bubble-up method
c. By breaking the input into smaller parts and merging them
d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **3**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes it easier to select items based on their relationship to the rest of the items
b. It makes the data unsorted
c. It slows down the process
d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **4**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It does not require recursion
c. It is based on the divide-and-conquer approach
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It compares adjacent elements and swaps them if necessary
b. It builds a sorted array one element at a time
c. It repeatedly finds the minimum element and moves it to the sorted part of the list
d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **6**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Deleting data from a list
c. Searching for data in a list
d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Merge Sort
c. Bubble Sort
d. Binary Search

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching
b. Sorting
c. Complexity
d. Merging

The correct answer is:
Complexity

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting
b. The parts are ignored
c. Each part is sorted independently
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Merge sort
c. Quick sort
d. Selection sort

The correct answer is:
Merge sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sorted()
b. order()
c. sort()
d. arrange()

The correct answer is: sorted()

Question **12**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort
b. Heap sort
c. Bubble sort
d. Insertion sort

The correct answer is:
Insertion sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It is less efficient than custom sorting algorithms
c. It requires external libraries
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Dividing the input into parts, solving each part, and combining the solutions
b. Sorting data sequentially
c. Rearranging data without sorting
d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Insertion Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort